The function $f(x) = (x-3)^2 + \frac{1}{2}$ has domain $\mathrm{D}_f : (-\infty, \infty)$ and range $\mathrm{R}_f : \left[\frac{1}{3}, \infty\right)$.

$\lim\limits_{x \to a^-} f(x)$ (or $\lim_{x \to a}$).

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = f'(a)$$

$$\int \sin x \, \mathrm{d}x = -\cos x + C$$

You can use $\int_a^b$ or $\int\limits_a^b$, and you can make them more bigger $\displaystyle\int_a^b$ or $\displaystyle\int\limits_{2a}^b dx = \left[\frac{x^3}{3}\right]_a^b$

$$\sum_{n=1}^{\infty} ar^n = a + ar + ar^2 + \cdots + ar^n$$

$$\int_a^b f(x)\, dx = \lim_{x \to \infty} \sum_{k=1}^{n} f(x_k) \cdot \Delta x$$

$$\vec{v} = v_1 \vec{i} + v_2 \vec{j} = (v_1, v_2)$$